

# Computing a Large Number of Independent Witness Points w.r.t Vertex Guards in a Simple Polygon

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# Problem

- ❖ Given a simple polygon in the form of vertices in  $(x,y)$  coordinates
- ❖ Find a “large” set of independent witness points with respect to vertex guards
- ❖ A vertex “sees” or is visible to another vertex if the line segment between them would be inside the polygon and not intersect any edges or points
- ❖ Witness points are independent if the sets of vertices they see are disjoint (i.e. no two witnesses see the same vertex)
- ❖ In this case, the set of independent witnesses is simply a subset of the polygon’s vertices

# What does “large” mean?

- ❖ For this problem, we want to maximize the number of independent witnesses
- ❖ However, computing the maximum (or the largest possible) set of independent witnesses in a polygon is very difficult to do in polynomial time
- ❖ The next best thing is a *maximal* set, which is a set of independent witnesses, such that it is impossible to add another independent witness to the set
- ❖ A maximal set may also be the maximum, but not necessarily
- ❖ For our purposes, a maximal set will be considered sufficiently large

# My Approach

1. Compute the visibility set (the set of vertices that a vertex sees) for each vertex
  - a. For each pair, use the Diagonalie test to make sure that the line segment between them forms an internal or external diagonal (it doesn't intersect any of the edges or vertices of the polygon)
  - b. Then use the InCone test to make sure that the line segment lies within the "cone" formed by the adjacent vertices (basically it's an internal, not external, diagonal)
2. Sort the vertices based on the sizes of their visibility sets (increasing order)
3. Iterate through the ordered set and eliminate vertices until you're left with independent witnesses
  - a. The first vertex A in the ordered set is made a witness. Suppose it sees vertices B and C. Then eliminate any other vertex that sees B and same for C. Then go to the next remaining vertex (potential witness), and repeat until you reach the end of the set.
  - b. The resulting subset of vertices is necessarily a maximal set of independent witnesses

# Using Symmetry to Reduce Running Time

- ❖ A naive approach for computing the visibility set for each vertex would be to test all pairs of  $n$  vertices in the polygon, giving an  $O(n^2)$  running time
- ❖ However, this is unnecessary as visibility is a symmetric relation
- ❖ If vertex  $A$  sees vertex  $B$ , then vertex  $B$  also sees vertex  $A$
- ❖ So once you found that vertex  $A$  sees vertex  $B$ , you can update the visibility set for both vertices so you don't have to test twice
- ❖ This gives a  $O(n(n+1)/2)$  or  $O((n^2 + n)/2)$  running time, which is an improvement

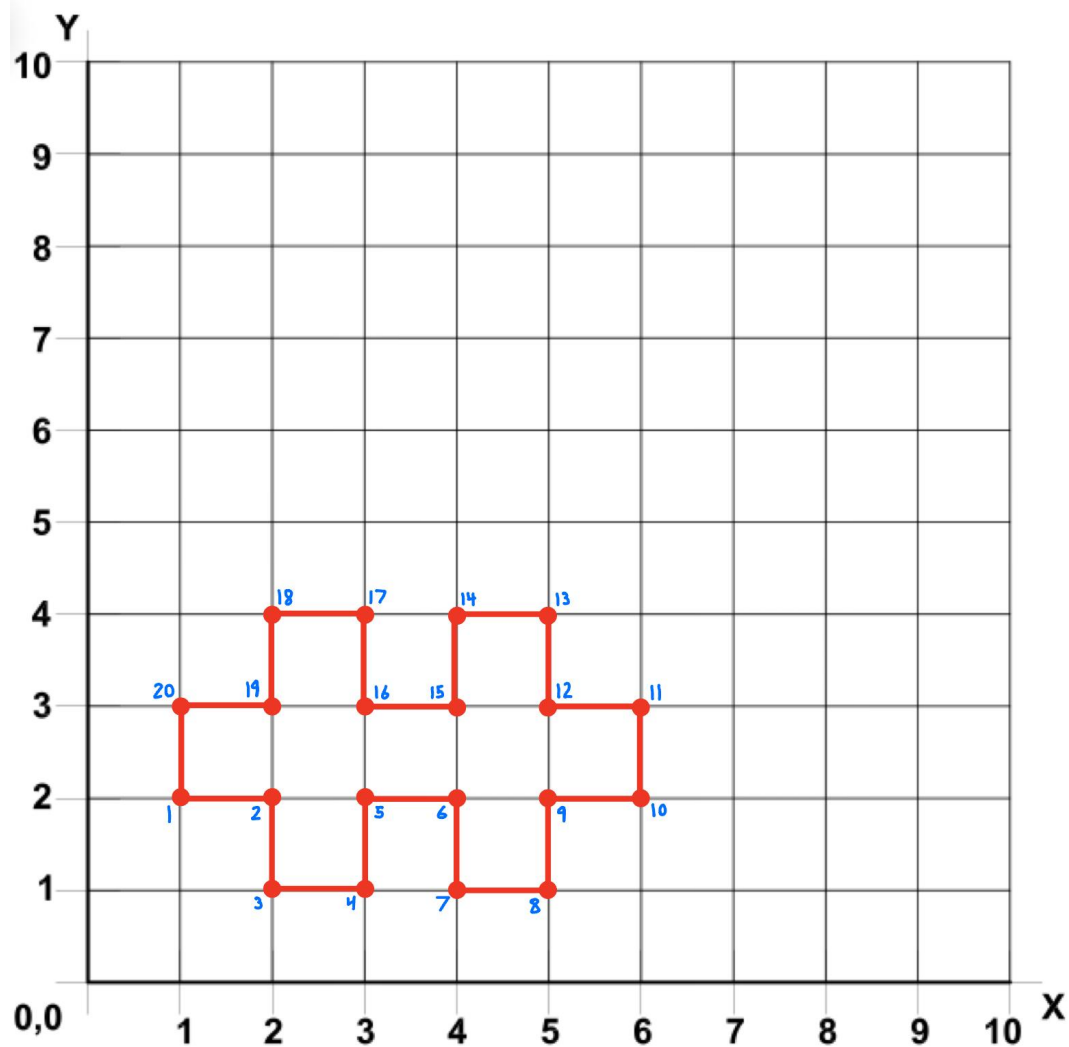
# Analysis of Time Complexity

1. Compute the visibility set for each vertex (using symmetry):  $O((n^2 + n)/2) \times \dots$ 
  - a. For each pair, use the Diagonal test:  $O(n) + \dots$
  - b. Then use the InCone test:  $O(1)$
2. Sort the vertices:  $+ O(n \log n)$
3. Iterate through the ordered set and eliminate vertices until you're left with independent witnesses:  $+ O(n)$

Total time complexity:  $O((n^3 + n^2)/2)$  which is strictly  $< O(n^3)$  for  $n \geq 2$

# Example 1 Input

$(1, 2), (2, 2), (2, 1), (3, 1), (3, 2),$   
 $(4, 2), (4, 1), (5, 1), (5, 2), (6, 2),$   
 $(6, 3), (5, 3), (5, 4), (4, 4), (4, 3),$   
 $(3, 3), (3, 4), (2, 4), (2, 3), (1, 3)$

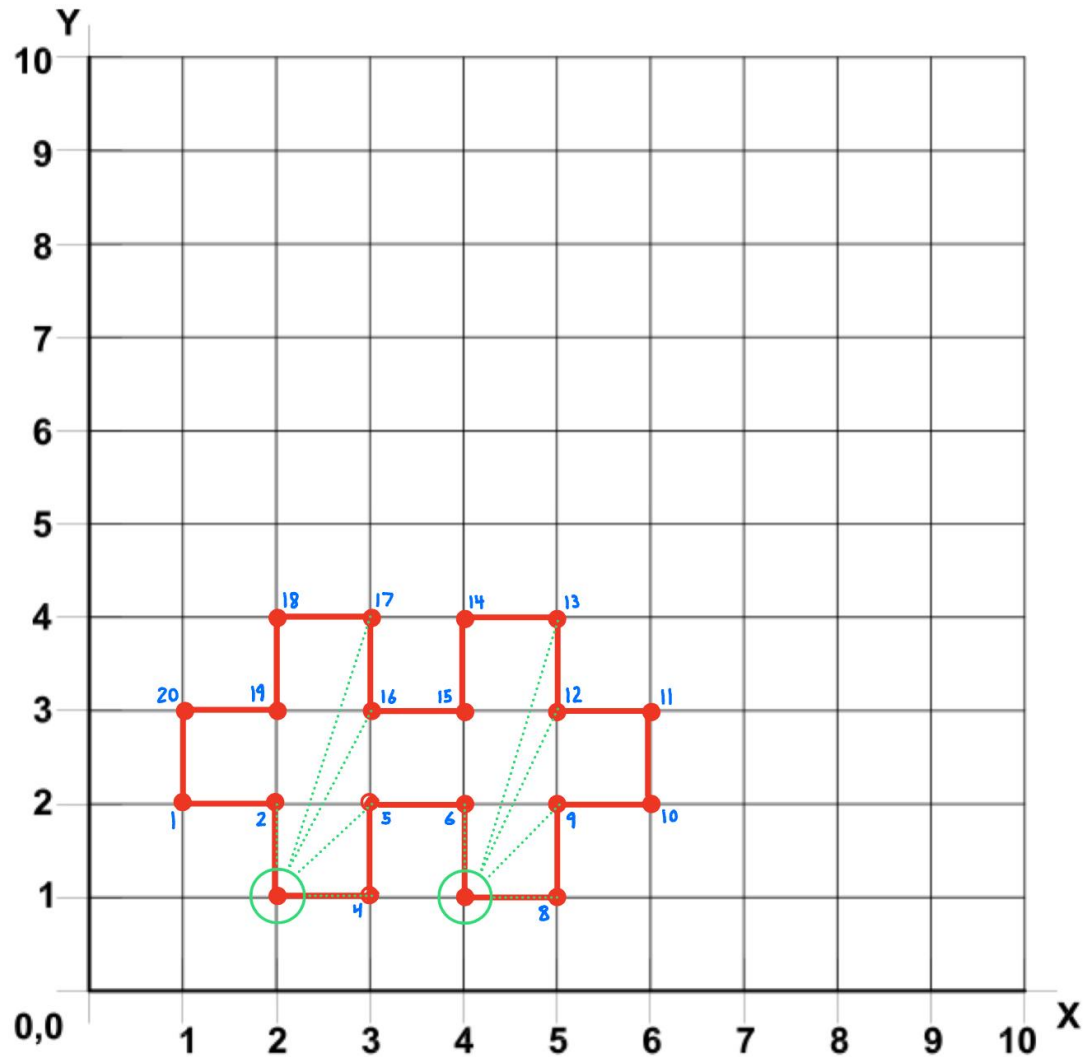


# Example 1 Output

Independent Witnesses:

Vertex 3 (2,1): 2 4 5 16 17

Vertex 7 (4,1): 6 8 9 12 13



## Example 2 Input

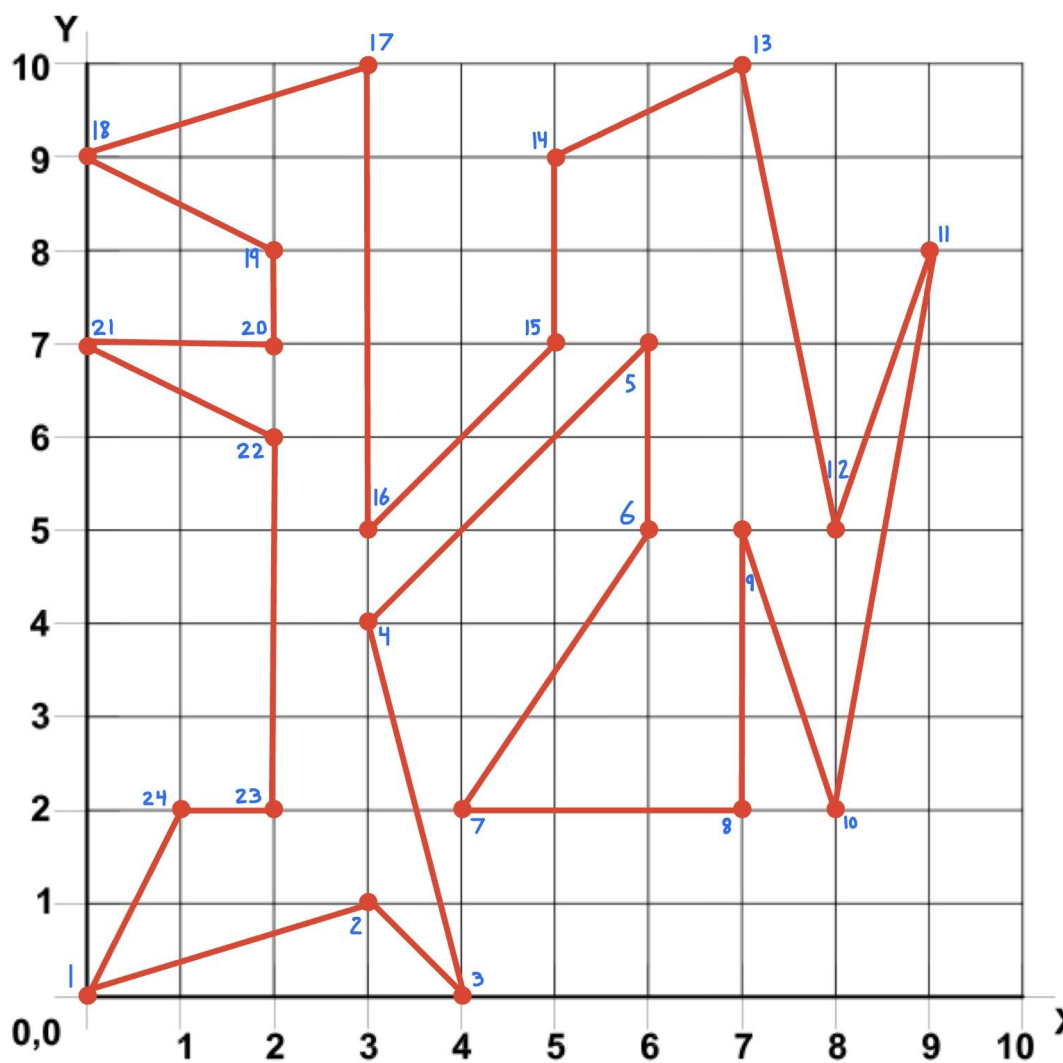
$(0,0), (3,1), (4,0), (3,4), (6,7),$

$(6,5), (4,2), (7,2), (7,5), (8,2),$

$(9,8), (8,5), (7,10), (5,9), (5,7),$

$(3,5), (3,10), (0,9), (2,8), (2,7),$

$(0,7), (2,6), (2,2), (1,2)$



# Example 2 Output

Independent Witnesses:

Vertex 11 (9,8): 10 12

Vertex 18 (0,9): 17 19

Vertex 21 (0,7): 20 22

Vertex 1 (0,0): 2 23 24

Vertex 7 (4,2): 6 8 9

Vertex 15 (5,7): 4 5 13 14 16

